

Decision Making Insights Of Self-Driving Cars

A lot of people are talking about autonomous vehicles, which use highly advanced and futuristic technologies. And although these come with sophisticated intelligence, their performance largely depends on decision making skills of humans on roads

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Intelligence of autonomous vehicles is based on nothing but simple robotics. So, if you have to decide on how to drive a vehicle well, you have to make decisions exactly how self-driving cars do it. Technologies like those mentioned below give strength to self-driving cars.

Deep learning vision. The cameras and the LiDARs in the front of the vehicle should be able to see the signboard that specifies the speed limit and the diversion ahead. Unlike humans, self-driving cars make strict decisions concerning traffic light rules.

Planning. This is responsible for all decision-making based on complete information about the surroundings.

Localisation and mapping. Nowadays, apps like Google Maps facilitate navigation at a global level. This module helps a vehicle in knowing its present location at a local level with respect to the nearby surroundings.

Localisation

One can use GPS to pinpoint location, but its accuracy is only up to a few metres at present. If you are deciding to stop because of a traffic light in front, or considering to overtake another vehicle ahead, then this kind of accuracy is not good enough. So, a centimetre-grade positional accuracy using a better localisation system eases the decision making process for a self-driving vehicle (with regards to where it is currently located). One can find out about the surrounding features using a feature-extraction algorithm.

The two images of the same place in Fig. 1 have been taken from a stereo camera at different intervals of time. From the second image (on the right-hand side), we can see that the trees and railing appear a little closer. This helps to calculate how much you have moved by integrating this information along with the time taken to find out your precise location.

However, stereo cameras are extremely noisy. So, the depth accuracy is pretty bad. Therefore, LiDARs provide a static 3D visual, wherein multiple images along with time are stitched together to give a final panoramic image (map).

The image in Fig. 2 is a visual representation of the IIIT Alla-



Fig. 1: Two images of the same place taken from a stereo camera at different intervals of time